

WASP-163b

R-2091

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_260402 · created 2026-08-10T00:55:04+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461133.009 +/- 0.057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.042
Transit depth (Rp/Rs)^2	2.2 +/- 1.25 [%]
Orbital Inclination (inc)	81.61 +/- 2.72
Airmass coefficient 1 (a1)	0.813 +/- 0.012
Airmass coefficient 2 (a2)	0.143 +/- 0.079
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	#1 - [440, 434]
Optimal Aperture	2.87
Optimal Annulus	4.74
Transit Duration (day)	0.031 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.042 vs 0.119 (deviation 0.69σ)
Duration fit vs published	0.031 d vs 0.0942 d (deviation 2.11σ)
Mid-time O–C	-2.1 min
Depth SNR	3.5
Residual scatter	1.54 %

Light curve

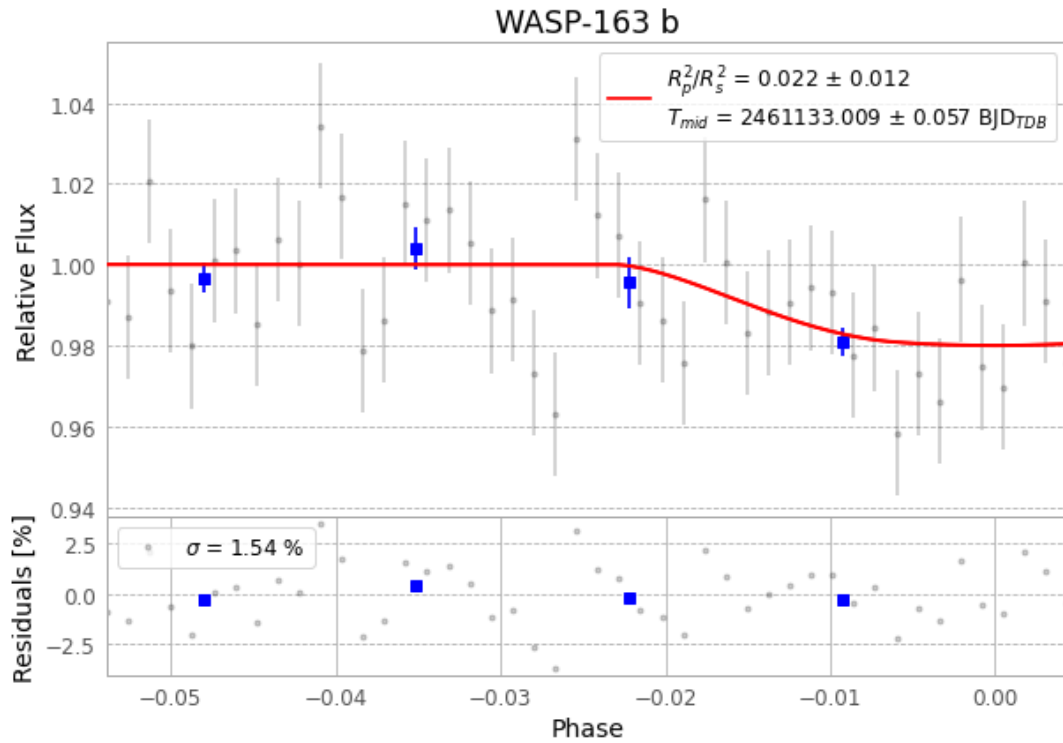


Figure 1 — FinalLightCurve_WASP-163 b_2026-04-02.png (SHA-256 4d0a721c7ccc22ce...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 168], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9c5372bb26b9e3a7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

48 FITS frames (32 MB), WASP-163260402100315.FITS → WASP-163260402123020.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-260402.log (8ca08a6f8673...), FinalLightCurve_WASP-163 b_2026-04-02.png (4d0a721c7ccc...), FinalLightCurve_WASP-163 b_2026-04-02.csv (1f5fea64a784...)

Compute resources

Wall time 111.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:55Z.